

MID TERM EXAMINATION
B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)
Third Semester, November 2022

Paper Code: CIC-209

Subject: Data Structures

Time: 1½ Hrs.

Max. Marks: 30

Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.

Q. No.	Question	Max. Marks	CO(s)
1. (a)	Briefly explain the Abstract Data Type (ADT) model with the help of a diagram.	2	CO1
(b)	Why are data structures required? List and explain various types of primitive and non-primitive data structures.	2	CO1
(c)	What is a sparse matrix? Explain briefly the 3-tuple (triplet) representation of a sparse matrix with an example.	2	CO2
(d)	Differentiate between static implementation (using arrays) and dynamic implementation (using pointers) of a list data structure.	2	CO2
(e)	Define the terms depth , height , internal node , and leaf node with respect to binary trees with the help of an example.	2	CO2
2. (a)	Write complete algorithms for: (i) Insertion of an element into a sorted singly linked list. (ii) Deletion of an element from a sorted singly linked list.	5	CO2
(b)	Consider the following traversals of a binary tree: Preorder: A B D E C F Inorder: D B E A F C Construct the unique binary tree based on the given traversals and show all step-by-step intermediate structures.	5	CO2
3. (a)	What is a Binary Search Tree (BST)? Write recursive algorithms for Preorder, Inorder, and Postorder traversals of a BST.	5	CO2
(b)	Why is a circular queue preferred over a simple linear queue? Demonstrate and explain the circular queue mechanism with the help of a suitable example.	5	CO2
4. (a)	Convert the following infix expression into a postfix expression using an operator stack: A + B * C - D / (E * F) ^ G Show the stack contents and output string at each step.	5	CO2
(b)	Construct a Binary Search Tree (BST) by inserting the following data elements sequentially: 35, 70, 50, 10, 48, 80, 55, 11, 85 Show the resulting tree after the insertion of each data element.	5	CO2

**** End of Question Paper ****